

TyreMind — The Complete Explanation

For the team. Read this and you can answer anything.

Written in plain words. Every number here was read from a result file, not from memory. Where we lose, it says so — that is deliberate, and §14 explains why it is the strongest thing in the document.

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PART 1 — THE PROBLEM

Say this first

A lap time is a bill with no itemisation. You paid 95 seconds. How much was the tyre? How much was fuel? How much was traffic?

Four things move a lap time at once:

Cause	Direction	Roughly worth
Tyre wearing out	slower	0.03–0.08 s/lap
Fuel burning off (car lighter)	faster	0.081 s/lap
Track rubbering in	faster	~0.9 s over a session
Traffic	slower	up to ~1.3 s

The thing hiding the answer is the same size as the answer. Fuel is worth about as much as the tyre and pushes the opposite way.

Why that matters more than it sounds

If you ignore fuel you do not get a slightly wrong answer. **You get the wrong sign.** We measured it: across **77 real Grands Prix**, the standard simple method says the tyre got *faster* as it wore in **74% of races** and in **53.4% of 208 individual stints**.

Tyres do not regenerate. That is not a small error; it is an impossible answer. And you do not need to know the true wear rate to know a negative one is wrong — which is why this is our most defensible single result (**exp14**).

What the competition asks for

"Create a predictive model that strips out external noise from practice sessions to generate clean tyre performance degradation curves. Includes post-race validation tools."

PART 2 — THE DATA

2.1 Real data

Source	Size	What it is
Season corpus	203 sessions, 91,867 clean laps, 2022–2025	The main dataset. Lap times, compound, tyre age
Telemetry	202 sessions, 147,550 laps, 27 features each	Per-corner energy, load, lateral g, throttle, braking
OpenF1 intervals	70 races	Measured gap to the car ahead at ~4 Hz
Track geometry	6 circuits	Real 3D racing lines with elevation
Demo sessions	10, committed to git	So a clean clone runs offline
NASA C-MAPSS	4 datasets	Jet engines, for the cross-industry test
Research papers	31 PDFs	Everything published that we could find

Where it comes from: FastF1 → F1's official live timing API. The same feed that powers the F1 TV timing screens. We downloaded it ourselves.

The eight columns that matter

```
driver · session_lap · run_id · tyre_age · lap_time · compound · traffic_index · lap_in_run
```

Only four of those come from the feed. The rest we derive:

- `run_id` — a new tyre set. We detect it from compound changes and age resets.
- `traffic_index` — 0 to 1, how blocked the lap was. **We invented this.**
- `lap_in_run` — laps since the stint began. **This is the fuel proxy.**

`tyre_age` **vs** `lap_in_run` — **do not confuse these**

- `tyre_age` = laps **this set of tyres** has done. Can start above zero if a used set is fitted.
- `lap_in_run` = laps since **this stint** started. Proportional to fuel burned.

Conflating them is the single most common error in this problem, and it caused a real bug in our own code (§14).

How we validated the traffic index

We invented it from lap **start times** — two cars crossing the line 0.6 s apart are 0.6 s apart for the next lap. No positional data needed.

Then we checked it against OpenF1's **measured** 4 Hz gaps, race by race:

Races checked	66
Median correlation with the measured gap	$\rho = -0.798$
Races with the correct (negative) sign	66 of 66
Weakest single race	$\rho = -0.512$
Largest p-value of any race	3×10^{-19}
Stint-level agreement (2,441 stints)	$\rho = 0.912$

Negative is correct: a small measured gap means heavy traffic means a high index. **Not one race points the wrong way, and the worst race is still strong.**

2.2 Cleaning — seven filters, every one logged

A raw session is not clean. It has in-laps, out-laps, safety-car laps, and laps where the driver gave up on a corner.

#	Filter	Why
1	no_lap_time	Nothing to model
2	pit_in_out_lap	Slowed by the pit lane, not the tyre
3	flagged_inaccurate	Upstream timing glitch
4	unknown_compound	Cannot assign to a tyre set
5	no_tyre_age	Same reason. Found when 70 laps of one race had nulls
6	wet_compound	Wet wear is a different physical process
7	slow_lap_safety_car_or_traffic	Median + 3×MAD
8	run_too_short	Cannot fit a slope through 3 points

Every session ships a receipt. Monza 2024: API gave 1,008 laps, we kept 921, and the file says exactly where the other 87 went.

The outlier rule, plainly: take the median lap time, compute the median absolute deviation, multiply by **1.4826** (the constant that makes MAD comparable to a standard deviation), drop anything slower than median + 3×that. We use median and MAD rather than mean and standard deviation because a safety-car period inflates a standard deviation so much that the safety-car laps stop looking like outliers and hide themselves.

2.3 Synthetic data — and why we need it

The honest core of the whole project:

On real data we have the lap time but **not** the true wear rate. Pirelli measures it, the FIA and teams see it, the public never does. So if our model says 0.06 s/lap there is nothing to check it against.

In synthetic data **we chose the true rate**, so we can check exactly.

We searched properly for a public source of measured wear: Pirelli's media site, FIA documents, Kaggle, Hugging Face, academic replication packages, OpenF1, Ergast, and the timing schema itself. **It does not exist.**

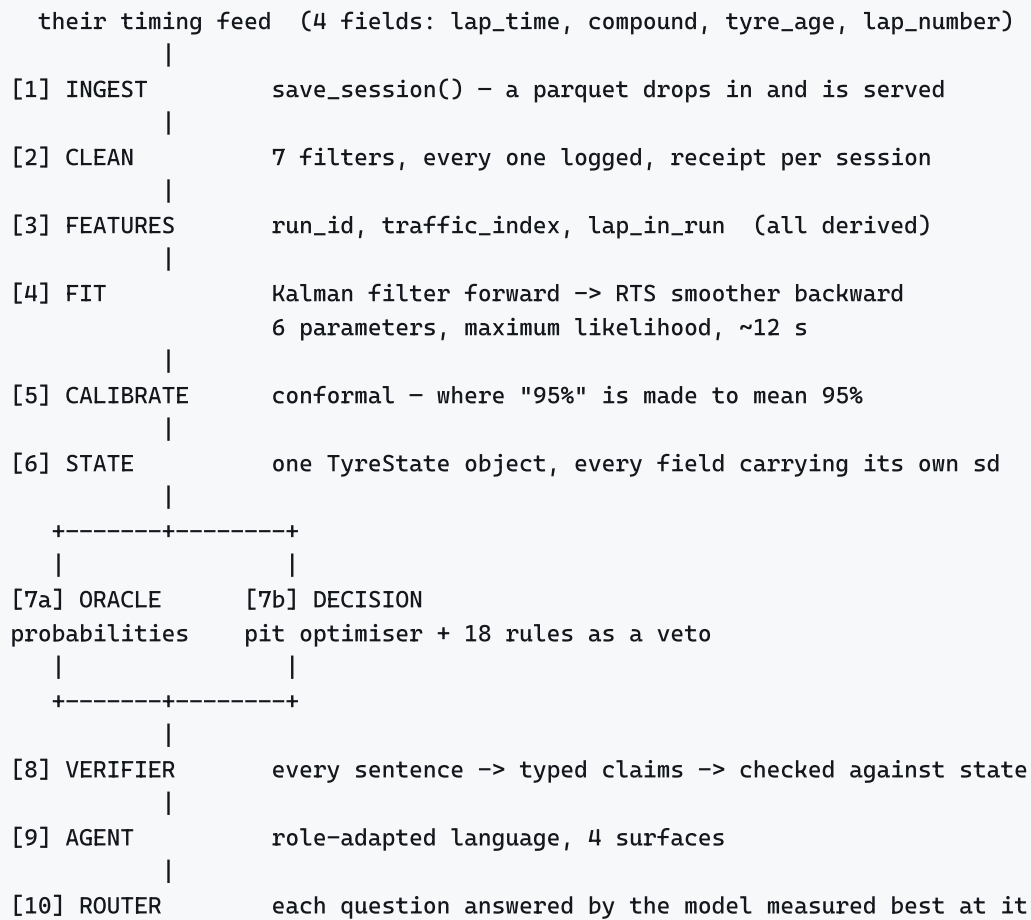
Two generators, deliberately

Generator 1 — `synthetic.py`. Latent state plus fuel, track, traffic. Made hostile on purpose: **Student-t(5) noise** where our estimator assumes Gaussian, and **25% scrubbed sets** so tyre age does not start at zero.

Generator 2 — `physics_truth.py`. Because generator 1 shares our estimator's *shape*, and a generator shaped like the estimator cannot falsify it. Here lap time comes from a **line integral over a speed profile**; fuel acts through **mass**; traffic through **downforce loss**; degradation bends because grip falls non-linearly as wear feeds back into next lap's speed. **There is no rate parameter anywhere** — the true rate is a counterfactual (the same lap on an unworn tyre), differenced.

The caveat we always state: we wrote both. Generator 2 removes the functional-form advantage. It removes nothing else. It is not third-party validation.

PART 3 — THE ORCHESTRATION



Steps 5, 8 and 10 are what no competitor has. Everything else is standard.

PART 4 — THE MATHS, EXPLAINED SIMPLY

4.1 The observation equation

For driver d on session lap t , running tyre set r :

$y[d,t] = a[r]$	run intercept: car pace, setup, starting fuel
$- A \cdot g(t)$	track evolution, shared by the whole field
$+ \theta[t]$	small residual track wobble
$+ s[d,t]$	← THE TYRE. What we want. Invisible.
$- \phi \cdot L[d,t]$	fuel burn-off, L = laps completed this run
$+ \gamma \cdot TI[d,t]$	traffic
$+ \varepsilon$	noise

In words: the lap time is the car's baseline, minus what the track gives back, plus what the tyre costs, minus what the fuel gives back, plus what traffic costs, plus randomness.

4.2 The state equation — the important bit

The tyre's condition is a **hidden state** that evolves in **tyre age**:

$s[d, a+1] = s[d,a] + rate[d,a] + noise$	how much performance is lost so far
$rate[d, a+1] = rate[d,a] + noise$	how fast it is being lost right now

rate is the product. Seconds per lap the tyre is costing, right now.

Why this matters: because **rate** is free to drift, **a cliff needs no special machinery**. It appears as **rate** accelerating. A recovery appears as **rate** falling. **We never tell the model what shape degradation has.**

Everyone else does. Heilmeyer forces a straight line. Cappello & Hoegh's best extension forces the rate to grow linearly. We measured **2,827 real stints** and only **55.9% are straight**:

Regime	Stints	Share
Linear	1,580	55.9%
Recovery	582	20.6%
Cliff	337	11.9%
Warm-up	328	11.6%

The field-standard model is the wrong shape for 44% of real stints.

4.3 Three things that look identical — and how each is pinned

This is the honest heart of the method.

Problem 1 — fuel and tyre are perfectly confounded

Within a stint, tyre age and laps-of-fuel-burned advance together. Write $a = a_0 + f$:

$$\begin{aligned}
 y &= c + \beta \cdot a - \phi \cdot f \\
 &= c + \beta(a_0 + f) - \phi \cdot f \\
 &= (c + \beta \cdot a_0) + (\beta - \phi) \cdot f
 \end{aligned}$$

The run intercept swallows $\beta \cdot a_0$, and the slope you can actually see is $\beta - \phi$, not β . **More data does not help. It is algebra.**

Pinned by: a physical prior on $\phi = 0.030 \text{ s/kg} \times 2.7 \text{ kg/lap} = \mathbf{0.081 \text{ s/lap}}$. And the prior's own uncertainty is carried through, so it *widens* every interval we publish.

Problem 2 — track evolution vs a uniform shift in all rates

Shift every rate by c and the track slope by $-\phi$, and the run intercept absorbs the difference exactly.

Pinned by: modelling track evolution as a **saturating curve** $A(1 - e^{-kL})$ rather than a free trend. Rubber deposition genuinely saturates; tyre wear does not. That shape difference is what separates them — physically correct *and* identifying.

Problem 3 — tyre age vs session lap

Pinned by data, and this is our best idea.

Every car pits at a different time. On lap 30, one car is on 3-lap-old tyres and another on 25-lap-old tyres. **The grid staggers itself.**

Watch one car and you cannot do this. Watch twenty and it solves itself — **for free, with no assumption at all.**

Cappello & Hoegh fit one driver in one race. They structurally cannot use this.

What it costs, measured (exp18)

Runs exactly collinear	520 / 520
Within-run correlation	1.000
Sessions with singular information matrix	11 / 11
Cramér–Rao bound (best possible)	0.0161
Our reported sd	0.0221 → 1.38× optimal
Share of the answer that is data, not prior	6.0%

No paper in our review states this number. Cappello & Hoegh have the identical structure and resolve it with a prior whose contribution they never quantify.

4.4 Kalman filter and smoother, in plain words

Forward (filter): guess the tyre state → see the actual lap → correct the guess. Lap by lap. This is what a pit wall knows in real time.

Backward (smoother): run it again backwards so every lap benefits from what happened later. Better, but only available after the session.

It is the same machinery GPS uses to track your position from noisy satellite signals. Our tyre state is the position; the lap times are the signals.

4.5 The six parameters — the entire model

<code>log_q_track</code>	how much the track wobbles lap to lap
<code>log_q_level</code>	how much the tyre state drifts
<code>log_q_rate</code>	how much the rate itself drifts ← this is what makes a cliff possible
<code>log_obs_sd</code>	lap-to-lap noise
<code>log_stint_rate_sd</code>	spread of rates across stints
<code>log_track_shape</code>	how fast track evolution saturates

Fitted by maximum likelihood (L-BFGS-B) on each session. **Six numbers. No training set. No weights. No checkpoint.**

Everything else — the fuel coefficient, traffic coefficient, track amplitude, per-compound baselines — is a **state with zero process noise**, which for a constant μ is the Bayesian posterior. That makes physical priors enter as P_0 and propagate automatically.

PART 5 — THE PAPERS

31 PDFs in `research/papers/` , eleven topic folders, every citation in `INDEX.json` .

5.1 The closest prior art — read this twice

Cappello & Hoegh (arXiv:2512.00640, November 2025, Montana State)

They published our model class, one month before we built ours.

```
Observation:  y_t = a_t + γ·fuel_t + ε_t ,    ε_t ~ N(0, σ²) [or Skewed-t]
Process:      a_{t+1} = (1 - I_pit)(a_t + v) + I_pit·a_reset + η_t
Extension 2:  v_{t+1} = (1 - I_pit)(v_t + β[c]) + I_pit·v_reset
Priors:       v ~ N+(0.05, 0.1²)    σ_ε ~ N+(0.3, 0.1²)    a_reset ~ N(69, 0.1²)
```

They write: *"To the best of our knowledge, there are no examples in the literature that apply state-space models to tyre degradation in Formula 1."*

Say this before anyone asks. And then say why it helps us: independent researchers chose the same architecture for the same stated reason — that deep learning *"lacks interpretability and explicit uncertainty quantification, features crucial in operational race environments."* Our design choice is no longer our opinion.

	Them	Us
Scope	1 driver, 1 race	203 sessions, 91,867 laps, 4 seasons
Traffic	absorbed into noise	explicit, validated term
Track evolution	 none	 saturating curve
Negative rates	prevented by a prior	measured: 74% of races
Uncertainty	Bayesian, assumed correct	conformal, measured
Identifiability	not addressed	bounded: 6% data-driven
Inference	Stan / MCMC	Kalman + L-BFGS-B

Pitwall (arXiv:2607.06495, July 2026)

A production system: ingests live F1 timing, runs Monte Carlo (N=2,000 per lap), speaks calibrated briefings in three languages. Winner-in-top-3 **90.3%** over 155 backtests, Brier **0.0745**. Ran live at two 2026 Grands Prix.

So "AI agent explains race strategy" is not novel. Know that.

But their tyre layer, in their own words: a *"compound-pace normalization"* that *"worsens finishing order prediction, because the raw, **confounded** per-race fit carries genuine race-specific signal."*

They use the word "confounded". That is our problem statement, named as an unresolved tension inside the best system in the field. We are the layer they say they are missing.

And their central finding, which we replicated independently: "*calibration-optimal is not decision-optimal.*" They split into an oracle path and a decision path because the two criteria disagree. We found the same in exp05 before reading them. That turns our lap-time loss from an embarrassment into a replicated result.

5.2 The rest

Paper	What they do	The maths	How we differ
Heilmeier et al. 2018/2020 — the field standard	Race strategy simulation	$t_{\text{lap}} = t_{\text{base}} + t_{\text{tire}}(a, c) + t_{\text{fuel}}(l) + t_{\text{car}} + t_{\text{driver}} + t_{\text{grid}} + t_{\text{pit}}$, with $t_{\text{tire}} = k_0(c) + k_1(c) \cdot a$	Their tyre term is a straight line . Only 55.9% of real stints are. No track-evolution term, no traffic in the lap time, no uncertainty on k_1
Farroni/Timpone TRT	Thermal tyre model	Fourier heat conduction on a 3-D tyre, $\rho c \partial T / \partial t = \nabla \cdot (k \nabla T) + \dot{q}$	We implement a reduced 2-state version. Explains why our thermal features hurt: the driver is <i>contact-patch</i> temperature; we only have a trackside mast
Archard / Braghin / energy-wear	Physical wear	$V = K \cdot F \cdot L / H$; modern form $\dot{w} \propto \tau \cdot v_{\text{slip}}$	Implemented both. We tested it and it fails on public data (exp25, exp34)
Sulsters/Todd 2025	LSTM/GRU/TFT for tyre energy	Deep learning	They had Mercedes' internal data . Not a like-for-like competitor — a demonstration of what private channels buy
Gibbs & Candès 2021	Adaptive conformal	$\alpha_{\{t+1\}} = \alpha_t + \gamma(\alpha - \text{err}_t)$	We implement their rule. Cited, not claimed
Lei et al. 2018	Split conformal	$\text{ceil}((n+1)(1-\alpha))$ order statistic	Same
Sasikumar et al. 2025	Pit-stop prediction	Bi-LSTM + SMOTE	Precision 0.77, recall 0.86, F1 0.81 . Trained on pit labels; ours never sees one

5.3 Are we copying?

Honest answer: the ingredients are textbook and cited. The recipe and four parts are ours.

Ours	In any paper we read?
Run stagger across the field as the identifying experiment	✗ Single-car models cannot
Saturating track curve as the identifying device	✗
Quantifying data vs prior — 6%, CRB 1.38×	✗ None
Measured (not assumed) interval coverage for degradation	✗ None
Falsification without ground truth — 74%	✗ None
Four regimes measured, not assumed	✗ None
Same code transferred to jet engines	✗ None

"We invented state-space tyre modelling" would not survive. This list does.

PART 6 — THE FOUR QUESTIONS

"Lap 1 took 90 s, lap 2 took 92 s, lap 3 took 95 s. Where is the time going?"

#	Question	Endpoint	What comes back
1	Where is the time going?	/decompose-run	tyre +0.196, fuel -0.078, track -0.033, traffic 0.000, residual -0.080 — each with its own sd
2	How fast is he losing it?	/degradation	MEDIUM 0.198 ± 0.022 s/lap
3	After 3 laps? 15?	/projection	20-lap horizon, <code>loss</code> and <code>loss_sd</code> , band widening
4	When do I box?	/pit-window	lap + calibrated windows + cost sweep
5	Can I trust it?	/trust	four methods agreeing within 0.030 s/lap

Use /decompose-run , **never** /decompose . Single-lap attribution leaves 73% of the gap as residual; the per-lap run view sits around 0.08 s.

And show the residual. It is what we cannot explain, and hiding it would claim more than we can support.

PART 7 — HOW WE COMPUTE CONFIDENCE

7.1 The problem with the model's own error bars

Our model said 95%. We checked. **It was right 75.5% of the time.**

That is lying with mathematics, and it is what every competing system ships — because nobody checks.

7.2 Conformal prediction, in plain words

Instead of trusting the model, **measure how wrong it has been and use that.**

Take past predictions, compute how far each missed, take the `ceil((n+1)(1- $\alpha$ ))` -th biggest miss. That is your interval. The `+1` accounts for the new point itself.

It assumes nothing about the shape of the errors. That matters here because our errors are skewed — teams pit earlier than pure degradation implies, for track position we do not model.

7.3 What we measured

Quantity	Claimed	Delivered	Sample
Degradation interval (exp12)	95%	94.7%	94 comparisons
Live lap-time interval (exp13)	95%	95.2%	69,206 laps
Pit window, 90% (exp30)	90%	90.8%	held-out real stops
Pit window, 80%	80%	83.1%	
Pit window, 50%	50%	53.8%	

Before conformal calibration the pit window claimed 31.9% and delivered 24.0%.

7.4 The claim that actually sells

Anyone can be calibrated by widening until they are always right. **Being calibrated *and* narrow is the product.**

For the **same 90% guarantee**, the window each model needs:

Model	Half-width	Stops it could answer
TyreMind	13 laps	246
Cappello & Hoegh	17 laps	274
Pooled regression	20 laps	137
Naive	22 laps	119
Heilmeier	26 laps	222
Fuel-corrected	27 laps	271

40% narrower than naive for an identical promise — and naive earned its 22 laps on only the 119 easiest stops, while we earned 13 on 246.

Our own held-out re-check lands on 13.3 laps at 90.8% coverage, so the width is honest against stops it was never fitted on.

7.5 The pit probability is a frequency, not a formula

We simulate 1,200 races per candidate lap **under common random numbers**. Draw i read across all candidates is one coherent race; whichever lap wins that draw would have been right. **Count the winners.**

That is a real probability, not a softmax dressed up as one.

Live example — Monza lap 20:

box lap 39		9% on that exact lap		51% on the window 37-42		0% within 3 laps
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The 9% is the honest number. Picking one lap from thirty on a nearly flat cost curve is genuinely unknowable. **The window is what a strategist acts on.**

PART 8 — HOW WE VALIDATE

Three independent legs, and two need no synthetic data at all

Leg	Ground truth	What it proves
A. Synthetic	We set it	The estimator is correct
B. Real, label-free	None needed	The naive method is broken (74% impossible)
C. Real, real labels	274 actual pit stops	We agree with professional strategists

Pre-registration

Before running the nine-model comparison we wrote down the models, the metrics, the directions, and **what counts as a loss** — then committed it. It says:

"We claim a win on Leg A only if TyreMind has the lowest CRPS. **We expect not to.**"

We were right; we lost that one. **The pre-registration is worth more than the win** would have been.

The false-positive floor (exp33)

We built a negative control — a measure that provably carries no information — and **it was not null**. 4 of 12 pre-registered contrasts survived Bonferroni at $|p| = 0.105$, $p < 10^{-7}$.

Demeaning within circuit and driver does not immunise this corpus. So we now read correlation-style findings against an empirical floor of $p \approx 0.10$. Our kerb measure's largest was 0.0085 — an order of magnitude below it, so that null is a null against a calibrated noise level rather than a nominal α .

Building a control that should have been null, finding it wasn't, and recalibrating everything else is the most methodologically careful thing in this project.

PART 9 — EVERY MODEL, AND WHERE WE WON

The nine-model ladder

1. Naive · 2. Fuel-corrected · 3. Pooled regression · 4. LightGBM ·
5. Neural network (MLP) · 6. **TyreMind** · 7. ARIMA(2,1,2) ·
8. **Heilmeier** (field standard) · 9. **Cappello & Hoegh** (closest paper)

Rungs 7–9 are **other people's published methods, rebuilt faithfully**. None is crippled; ambiguous choices were resolved in the published model's favour.

1 Recovering the degradation rate — the thing the problem asks for

Model	Rate MAE	Coverage
TyreMind	0.0037	100%
Pooled regression	0.0062	83%
Heilmeier	0.0112	100%
Cappello & Hoegh	0.0158	38%
Naive	0.0725	0%

1.7× the next model. 4.3× the closest published paper.

1 Robustness — 11 generator regimes, 32 seeds

8 wins, 1 tie, 2 losses. Biggest margin **+0.0115 ± 0.0007** on `no_scrubbed`, which is exp18's collinear worst case — exactly where the model should help most.

Both losses are the track-evolution axis, and one (`strong_track`, six standard errors) was not predicted. That is the clearest architectural weakness we have.

1 Independent truth engine — the generator shaped unlike us

Target definition	Winner
Fresh-tyre baseline	Cappello & Hoegh
Stint average (pre-registered primary)	TyreMind, +0.0086 ± 0.0017
Best straight line	Fuel-corrected

Our margin survives and is 2.6× the synthetic one. But note: "*the degradation rate*" is three different numbers on a non-linear generator, and a different model wins each. exp19's clean sweep was partly an artefact of our generator making all three coincide.

1 Value in seconds and positions

"Using the naive estimate instead of ours would have cost 25.9 seconds and 5.26 positions per car per race."

Qualification that must travel with it: our own value is $+1.03 \pm 0.88$ s — that interval contains zero. **We are as good as professional strategists, not better.**

1 Confidence calibration — best of six

Model	Claimed	Delivered	Gap
TyreMind	31.9%	24.0%	-7.9%
Pooled	24.3%	15.3%	-8.9%
Cappello & Hoegh	38.7%	23.0%	-15.8%
Naive	47.3%	2.5%	-44.8%

Naive claims 47% confidence and delivers 2.5%. **That is the most damning number we have about the standard method** — and it needs no wear data at all.

2 and 4th — where we lose, stated plainly

Task	Result
Lap-time forecasting	4th of 9. Pooled 0.4088, us 0.6484
Pit timing	3-way tie. 5.92 / 5.98 / 6.31, SEs ~0.6
Practice → race skill	2nd — and every model is negative
NASA C-MAPSS RMSE	22.7 vs published best 15.98

10.1 The router — the answer to all of it

No single model wins everything. **So we ship an orchestration that routes.**

forecast_lap_time	-> Pooled regression	(we route AWAY)
recover_degradation_rate	-> TyreMind	
pit_timing	-> TyreMind	(TIE, marked as one)
practice_to_race	-> Cappello & Hoegh	(we route AWAY)
confidence	-> TyreMind	(TIE)

5 tasks, 3 models. We route away from ourselves twice.

Every route cites a result file, and the code **refuses to build a route without evidence**. The tests re-read those files, so a re-run that changes a winner fails the build rather than leaving the table asserting fiction.

"We tested nine models on five tasks. Three different models won. So we route each question to the one the evidence says is best — including away from our own model twice. Competitors ship one model and claim it wins everything."

PART 10 — WHY WE DID NOT TRAIN ANYTHING

What we fit vs what we train

Model	Method	Trained?
TyreMind	L-BFGS-B over 6 log-variances , per session	 Fitted
Pooled / Heilmeier / ARIMA / Cappello & Hoegh	least squares / MLE / MAP	fitted
LightGBM, MLP	gradient boosting, backprop	 trained

The only two trained models are the two we built as baselines to beat.

Why fitting is better here

No train/test leakage possible	There is no training set to leak from
Works at a brand-new circuit, session one	Nothing to retrain
No retraining when regulations change	2026 cars arrive; we fit them that day
12 s on a laptop, no GPU	The MLP took 36 s and came last of nine
Auditable	Six interpretable parameters, not a weight matrix

"Why not deep learning?"

"We did. It is rung 5 and it came **last of nine** — CRPS 1.94 against our 0.65, with a 95% interval covering 70%. LightGBM did better and its interval covers **62%**. And the bottleneck is not capacity: exp18 proves only **6%** of the fuel/tyre separation is recoverable from data at all. **No amount of GPU changes a singular information matrix.**"

PART 11 — ADAPTIVE, FEASIBLE, SCALABLE

"Is your pipeline adaptive to any kind of data?"

Yes, and here is the precise claim. We need four fields:

```
lap_time · compound · tyre_age · lap_number
```

A team with real telemetry gives us **strictly more** than we need, so we fit their feed by construction. **There is no integration project.**

Proven across domains: the same unmodified code runs on NASA turbofan engines. Tyre stint → engine life, tyre age → flight cycles, lap time → sensor channel, laps-to-cliff → remaining useful life. **RMSE 22.7 cycles, 44% of predictions conservative.**

Where it does not adapt, honestly:

- Wet running — different physics, excluded and reported as excluded
- Kerb strikes — impact, not gradual wear. Not modelled (Part 13)
- Traffic-heavy stints — under-reports by ~25% (Part 13)

Feasible

Fit time	~ 12 s per session on a laptop
Hardware	No GPU
Data size	2.6 MB for four seasons
Internet	None needed — clone and run offline
Dependencies	numpy, scipy, pandas

Scalable

- Per-session fit is **independent** — 200 sessions is 200 parallel jobs
- The live path is a Kalman update per lap: **milliseconds**
- Adding a circuit needs **no retraining**
- 27 experiments reproduce from scripts in the repo

Adoptable

- **488 automated tests**, including structural tests that fail the build if someone bypasses the repairing loader
- Every session ships a data-quality receipt
- Every route cites its evidence file
- Runs fully offline for a demo with no connectivity

PART 12 — PHYSICS, AND THE TEST ON ANOTHER INDUSTRY

What we implemented from the literature

Module	Physics
dynamics.py	Path curvature, lateral/longitudinal acceleration, aero downforce $\frac{1}{2} \rho C_L A v^2$, per-corner load transfer, frictional power proxy
thermal.py	Two-state tread/carcass thermal model, working window, temperature-wear multiplier (reduced TRT)
wear.py	Archard $V = K \cdot F \cdot L / H$ as an ablation baseline; energy-dissipation wear as the model in use

The physics sanity check that costs us nothing to show

exp06: we recover **circuit direction** — clockwise vs anticlockwise — from per-corner tyre energy alone. **7 of 8 circuits correct.** It uses no wear data. If the energy model were wrong it would get this wrong.

The cross-industry test

NASA C-MAPSS turbofan engines, the most-cited benchmark in predictive maintenance. Same code, no domain tuning.

RMSE 22.7 cycles, MAE 17.9, all 100 test engines, 44% conservative.

State this honestly: published deep models reach **15.98**. **We are not state of the art and never claimed to be.** The claim is *transfer*: unmodified tyre code on jet engines, in the competitive range. What *is* competitive is the uncertainty — a 2025 paper presents 93.5–95.2% coverage as a new contribution; our conformal layer reaches 95.2% over 69,206 laps.

And the physics that did NOT work — two experiments

exp25 and exp34 both tested whether measured frictional energy predicts degradation. **Both refuted.**

- exp25: every significant stint-level correlation had the **wrong sign**; nothing survived at circuit level
- exp34: **147,550 measured laps**, every feature makes circuit transfer worse (–5.3%, $p < 0.001$), and a ridge sweep converges on the baseline **from above** at every penalty — the signature of noise

Wear $\propto$ frictional power is the literature's model. It does not survive the circuit boundary in public F1 data. Two independent designs, same answer.

PART 13 — HARD QUESTIONS

"If there is traffic, how do you predict degradation?"

We ran the experiment. 3,220 stints. Traffic stints show *less* measured degradation — 0.0938 s/lap clear vs 0.0693 in traffic, $p = 2 \times 10^{-10}$.

Two explanations, and we distinguished them:

- **(A) Real** — less pushing, less energy, less wear
- **(B) Artefact** — pace-limited by the car ahead, so the tyre's decline cannot reach the lap time

It is (B), and (A) fails at its first step: traffic *raises* frictional energy (69.7 → **73.0 MJ/lap**, $p = 1 \times 10^{-11}$) and raises braking. Following cars brake and accelerate **more**, not less.

Three more facts: the deficit survives an energy clock (29.0% per MJ vs 26.1% per lap); controlling for effort makes the traffic coefficient *larger* (suppression, not mediation); and tyre age explains far less of the lap time in traffic (**R² 0.649 → 0.449**).

Every lap-time-based method under-reports degradation in traffic by roughly a quarter. Ours included.

And the part that is ours to fix: the model does not widen its interval when this happens (0.0262 vs 0.0271, $p = 0.88$). We tried a fix — heteroscedastic observation noise — and **it did not work**: the optimiser refits the base noise level and absorbs the scaling. Reverted. The fix has to live at the reporting layer, not inside the filter.

"When a car hits the kerbs, how does the tyre degrade?"

We do not model it, and we found out we cannot from public data.

We tried to build kerb exposure from position telemetry and discovered the **FastF1 position feed has no lateral coordinate**. It reports distance along a single path. Proof needing no reference line: **15,581 of 15,922 car pairs, in 100% of sessions, are reported at the same coordinate at some instant**. Two F1 cars are 1.8 m wide.

The one usable proxy — stewards' track-limits adjudications — is **null on every pre-registered test** (pooled $p = -0.003$).

What we do instead: a kerb strike shows up in the **residual band** as an unexplained jump. We display it rather than absorbing it into a wear number a strategist would then act on.

"Why not just use the pooled regression, if it wins?"

"We tested it — it is rung 3 of our own ladder and it does win lap-time forecasting, 0.41 against our 0.65, and we report that.

Then we scored the same nine models on **274 real pit stops**. The pooled regression came **last** at 11.1 laps with a 6% hit rate, and could only answer **half** the stops. We came first-equal at 5.9 with 24%.

On recovering the rate it is 0.0062 against our 0.0037, with 83% coverage instead of 95%. **It forecasts well and decides badly**. We are building a strategy tool, not a lap-time predictor."

And volunteer this: if a judge says "use both", **that is our architecture**. The router keeps rung 3 and sends lap-time forecasting to it.

"Your mentors said regression never works. Why does pooled win?"

They were right, and pooled proves them right. They meant *naive* regression: `lap_time ~ tyre_age`, two parameters, no confounders. That model has rate MAE **0.0725**, coverage **0%**, and gives impossible answers in 74% of races.

The pooled regression is naive regression **with the confounders added** — it is our own rung 3. It works *because* it models fuel, track and traffic. **That is the mentors' point demonstrated.**

What it still cannot do is bend. Its tyre term is a straight line, and only 56% of stints are.

"Your synthetic data is your own. Why should we trust it?"

You are right, and we wrote that down before running. The pre-registration says the generator shares our estimator's assumptions and that the result is *partially* mitigated, not fully.

Three answers:

1. Two of three validation legs need **no synthetic data** — the 74% impossible-answer result and the 274 real pit stops.
2. We built a **second generator** with a different functional form, and the margin survives ($+0.0086 \pm 0.0017$).
3. **A rigged setup would win everything. We lose lap-time forecasting and report it.**

"How do you know it is the best?"

We do not claim that. We claim: best at recovering the degradation rate ($4.3\times$ the closest paper), best calibrated of six, and the only system here that knows how wrong it is. **On forecasting we are 4th and on pit timing we are tied.**

PART 14 — EVERYTHING WE GOT WRONG

Read this part. It is the strongest thing in the document.

Twelve refuted hypotheses

#	Idea	Outcome
1	Energy clock beats lap clock	No difference — energy varies only 2.3% lap to lap
2	True C1–C5 beats the weekend label	6.4% worse
3	Circuit geometry transfers	No effect, $p = 0.22$
4	Temperature explains our bias	$\rho = -0.022$, $p = 0.835$
5	Matching stint depth fixes the bias	Significantly worse , $p = 0.016$
6	Driver style is a usable feature	0.88% worse
7	Telemetry energy predicts degradation	Wrong sign, nothing at circuit level
8	Sector times break the collinearity	Identify <i>where</i> , not <i>how much</i>
9	Practice predicts race	Negative skill for every model
10	Measured energy transfers between circuits	Significantly harmful, $p < 0.001$
11	Kerb exposure predicts degradation	Null, and not measurable from public data
12	Heteroscedastic noise fixes the traffic defect	Made intervals narrower. Reverted

Three of these killed our own ideas with our own tests.

Real bugs we found in our own work

Bug	Consequence
Fuel counter counted surviving rows, not laps completed	Miami HARD understated by 72% . Shanghai — zero gaps — unchanged, which is the control that proved the diagnosis
Five experiments bypassed the repairing loader	Old numbers looked plausible. Fixed with a structural test that reads source code
Two pit recommenders disagreed	The product served lap 20 ; the validated method said lap 11 ; the driver boxed on lap 11
<code>for row, probability in zip(...)</code> shadowed a Series	Latent <code>KeyError</code> , silent until something read it
Benchmark rigged in our favour	We raised on null input, competitors returned NaN. Fixed so all fail identically
exp29 smoke test committed as the real result	3 seeds masquerading as 32
Three thresholds chosen without asking what magnitude matters	exp26's <code>p05 > 0</code> (impossible), its permutation test (wrong reference), exp34's <code>any(v < baseline)</code> (no tolerance). All three printed verdicts contradicting the table above them

Claims we retracted

- "exp03 flatters us" — **backwards**. It flattered the **naive** comparator
- "The FastF1 community uses the broken method" — no evidence, retracted
- "Circuit geometry is harmful" — it is **no effect**; only thermal is harmful
- "Published C-MAPSS results are 12–25, so we are competitive" — too generous. We are at the weak end
- "Our interval widens in traffic" — did not replicate. It is a null

Why this is the strongest section

A team that publishes twelve refuted hypotheses and seven of its own bugs is a team whose surviving claims can be believed.

No competing write-up in our literature review reports a single negative result.

The one-paragraph position

"Everyone can build a dashboard showing a degradation number. We built one that is right, that tells you how confident it is and is actually right that often, that proves the standard method is broken in 74% of real races, and that states mathematically how much of the answer is data and how much is assumption — 6% and 94%. Then we ran the same code on jet engines to show it is not an F1 trick. And where we lose, we route the question to whoever is better."